

Unity requires physically enforced agreement

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Abstract

A conscious scene is experienced as one scene, although many brain regions contribute to it. We propose that this unity requires physically enforced agreement: the regions must be coupled so that any change in one, as long as it exceeds that region's own noise, reaches the others and pulls them back toward agreement within the time a percept takes to form. This is a hypothesis about how unity is physically realized, and it is stricter than ordinary cause and effect, since the bits of a computer also act on one another. The condition is tested where each carrier of content is read. Machine-checked results make the digital case precise: under noise, content read from bits responds to a small physical change by at most the probability that a bit comes near its threshold, a bound that is attained, so a digital system whose bits are read well clear of their thresholds fails the condition. Applying one test to both substrates, the chance that any of 10 independent model membranes fitted to recordings from awake mice reaches a sensitive state within a percept's time is at least 0.91, while the value a clocked latch reads never does. We take on the hardest case, a fast digital replacement whose resolution is finer than biological noise, and propose an experiment that reconnects two cortical regions through an analog

or a quantized loop, varies bandwidth and quantization separately, and compares the outcome with a model in which only bandwidth, delay and nonlinearity matter. Reports of unity enter through an explicit assumption linking them to experience. Simulations show that the two loops can be matched on the information they carry; the coupling and experiential tests need the proposed preparation. The condition also predicts how fast and how far cortical coupling must restore agreement.

1 Introduction

When you watch a bird fly across the sky, you see one bird in one scene. Yet many parts of your brain work on that scene at once. A visual area tracks where the bird is, and a parietal area tracks the same location to plan a reach toward it. Each part describes a piece of the same situation, and their descriptions agree. This paper asks what kind of agreement that has to be.

Parts can agree in two ways. Picture first a set of weights joined by springs. Each weight's position is a graded physical quantity that pulls on its neighbours all the time, so no weight can move, even slightly, without dragging the others after it. The weights are coordinates of one physical state, and above their own noise they cannot vary independently at any scale. We call this agreement *physically enforced*. Picture next a set of account books kept consistent by a clerk, who at fixed times compares the books and corrects any mismatch. Between checks a smudge on one page changes nothing in any other book. The books agree only because a rule makes them agree, and we call this agreement *computed*. The springs make one system in the sense physics uses; the books are many systems kept in step by a protocol. Both involve causes, of course: the clerk reads one book and writes in another. The difference is which changes get through. Any displacement of a weight changes the pull on its neighbours, whereas a smudge that leaves a numeral readable changes nothing the clerk does.

A digital chip is of the second kind. Each bit is a separate record, built to ignore small disturbances: a voltage a little above or below the level for 1 still reads as 1. That makes computers reliable, and it also seals each bit off from small changes in every other bit, so whatever agreement a program produces among its bits is computed. The line falls where the bits are read: a converter that samples a graded signal finer than its noise is judged by the same test, and can pass it. Our premise is that the unity of an experience is agreement of the first kind.

Premise P (physical unity). A system's contents are unified only if, for every two regions whose contents overlap and in every window of the contents' own time scale, some content-bearing carrier of each region responds gradedly, above noise, to a carrier of the other, and the coupling between them shrinks their disagreement within that window. Different carriers and moments may serve the two directions and different overlaps.

In short, the regions that contribute to one experience must be held together by a physical coupling that feels small changes above noise and pulls disagreements back, and not by an exchange of symbols. P is proposed as a necessary condition for unity. It concerns how conscious contents are held together; which contents are conscious, and what they are like, lie outside it. It says nothing about where in the brain unified contents live: it constrains whichever regions carry them, whether they span the cortex or are confined to a posterior zone of it (Koch et al., 2016).

A finer grain of function. P is still a functional condition: it picks out unity by what physical parts do to one another. What sets it apart is the level of detail, or *grain*, at which it describes them. Functionalism leaves the grain open. Described coarsely enough, a nation whose citizens exchange messages by radio could run a brain's program (Block, 1978), while the biology of cognition has been argued to tie mental states to a finer grain (Godfrey-Smith, 2016). Grain includes time: a radio nation standing in for a brain

would run the right program at the wrong pace, still settling where the bird was after it had flown, and P's content window excludes it on that ground alone. P sits at the grain of graded physical dynamics, and computational functionalism about unity at the grain of the algorithm. Arguments that consciousness depends on embodiment, biology or the substrate often leave open which physical property does the work. P names one and states it as a mechanism: what a system's own readers take in, through what margin, against what noise and within what window. A substrate requirement then becomes something to measure, by one protocol in cortex and in chips, and to test against a specified rival model. For cortex the two grains agree. They come apart for a digital simulation of cortex: it can reproduce the course in which the contents agree, while every bit inside it stays well clear of its threshold, so that small physical changes to one bit reach no other. The experiment of Section 6 asks which grain unity tracks. It reconnects two brain regions through a graded or a digital link, varies bandwidth and quantization separately, and tests an explicit model in which only bandwidth, delay and nonlinearity matter.

The argument. The case for P has three parts, defended in Section 3.

- *Wholeness*, an account of unity: unified experiences are parts of one total experience. A physical account should therefore say what physically makes them belong together.
- *Counting*, our hypothesis about what does: regions belong to one physical whole when chains of mutual responses above noise join the carriers of their contents. For a direct link, every change of a carrier between its noise and its threshold must reach the other.
- *Restoration*: the coupling pulls overlapping contents back into agreement before those contents change.

Counting says what the physical whole is; restoration says which couplings keep its contents consistent. Together they motivate P. The formal results then show what follows: noise margins cap how much a record can respond, records drop out of the chains that counting builds, and graded readers can join regions. Whether experience tracks this condition is argued in Section 3, and tested in Section 6 through a stated link between experience and reports.

Noise and time. Two refinements make P applicable to noisy carriers. Physical systems are noisy, so P is read on probabilities: a change counts only if it is larger than the part’s own noise, and it must then move what the part does next by at least as much as the part fluctuates by itself. And a neuron is sensitive to small inputs mainly when it is close to firing, so asking every neuron to be sensitive at every instant would count no network of spiking cells as one system. P asks instead that *some* carrier of each region register the other regions within each window of the content’s time scale.

The paper makes three claims:

1. P predicts that a bridge which restores unified experience between two cortical regions supplies graded coupling and restores agreement within the content’s window; a bridge certified to lack that coupling which still preserves a calibrated readout of unity challenges P and its readout assumption together. A specified dynamical model tests whether bandwidth, delay and quantization explain the readout instead (Section 6).
2. Under a noisy margin, a small physical change moves the next content’s distribution by at most the probability of coming near a threshold, a bound that is attained. Digital readouts held within their margins fail the response test, while membrane models, including fits to recordings from awake mice, reach sensitive states within a percept’s time scale (Section 3).

3. Enforced agreement must arrive before the content changes, so P predicts that cortical coupling relaxes within 166 ms across the cortex; on a noisy sheet of coupled oscillators, the coupling’s reach decides how far agreement in phase holds (Section 4).

The formal results listed in Appendix K are machine-checked in Lean 4. The link to experience is carried by the realization hypothesis and the readout assumption, and the tests are designed to put both at risk.

2 Unity as agreement between overlapping regions

Think of an atlas. Each page maps one patch of ground, neighbouring pages overlap, and where they overlap they show the same roads. Pages that agree on every overlap fit together into one map of the whole territory. We treat the brain’s regions the same way. Each describes part of a shared situation, and we propose that unity, on the side of content, is this agreement: local descriptions that match wherever they overlap. When they match, they fix a single description of the whole.

Synchrony is not unity: two regions can oscillate in phase while encoding a stimulus at two different locations. Unity is about what overlapping regions describe, and synchrony is at most one route to it.

Carriers and traces. A region’s *content* is a variable the system itself reads, not merely one an outside observer can decode: the regions it drives respond to it, and it determines the quantity the region represents, such as a location. A variable that meets both conditions is a *carrier*; one that only an observer can decode is a *trace*. The supply current of a digital circuit reveals enough about its logical state to recover a secret key (Kocher et al., 1999), yet no gate reads it, so it is a trace. In cortex, membrane potentials and spike

times are carriers, because the cells they drive integrate them. The same test says *when* a variable is a carrier: at the moments its readers take it in. Feed-forward inhibition confines a pyramidal cell’s integration of its inputs to a window of a few milliseconds (Pouille & Scanziani, 2001; Gabernet et al., 2005), short against the interval between the spikes it receives (König et al., 1996), and the timing of first spikes in somatosensory cortex carries information about a stimulus that the spike count does not (Panzeri et al., 2001). A clocked latch, by contrast, looks at its input only during a short aperture at the clock edge. The input’s value there is a carrier. The course of the voltage between edges, which nothing reads, is a trace, like the supply current. So a carrier’s *next content* is what its readers take from it next, at the time they take it: for a membrane, the time of its next spike; for a node that feeds a latch, its value at the next clock edge.

Agreement is measured between regions whose contents represent a shared variable. The regions, and each one’s content variable, are fixed by this causal role and by decodability before any agreement is measured. Nobody can make the measurement trivial by choosing the regions after the fact.

A running example. A visual area represents the bird’s location on a retinotopic map, and a parietal area represents the same location to plan a reach. Each area’s content is its estimate of the location, carried by the membrane potentials and spike times of the cells that encode it. Compatible content, here, is the two estimates agreeing.

The definition, stated precisely. Region i carries a content s_i , a description of the variables it represents, and regions i and j overlap in the variables both represent. Restricting a content to an overlap keeps only the shared variables. The contents are *compatible* when every two agree on their overlap, $s_i|_{i \cap j} = s_j|_{i \cap j}$. Compatible contents extend to exactly one global description whose restriction to each region is that region’s content. Compatibility is our proposed correlate of unity, and the *overlap discrepancy*

measures how far contents fall short of it: the largest distance between two contents restricted to a common overlap.

Approximate agreement still fixes one description: if every overlap discrepancy of real-valued contents is at most ε , a weighted average of the contents that cover each variable is within ε of every region’s content. Agreement is measured by decoding, and two accurate decoders agree whatever produced their agreement, since $|d_A - d_B| \leq |e_A| + |e_B|$ for errors e_A and e_B . The measure is therefore the agreement *in excess of* what the two errors imply. These results are proved in a companion paper (Bobadilla-Suarez, 2026), and P adds a condition on how that agreement is produced.

Two stages. The companion paper tests whether excess decoded agreement tracks unified content. The bridge of Section 6 then tests how that agreement is physically produced. Moving from either to experience needs a readout checked against reports of unity while visibility, attention and task performance are varied (Section 6). If decoded agreement fails the companion’s test, the experiential test falls back on that report readout. In animals, behaviour tests integration across regions and the mechanism behind it.

3 Enforced agreement and computed agreement

The argument in outline. Five definitions make “physically enforced” precise. Two theorems then show that content read from bits barely responds to small changes, first without noise and then with it. Next comes the case for the premise: wholeness asks for a physical realization, counting says what that realization is, and restoration adds a deadline. Finally, one measurement protocol, applied to a cortical membrane and to a logic node, shows where each substrate’s carriers pass.

Definitions. Each is stated in words first.

- *Micro state and content.* A system has a physical state, its *micro state* $x = (x_i)_i$, made up of the states x_i of its regions. One step of its physics carries x to $f(x)$. A *content map* π reads each region's content off the micro state. The content level is *closed* when the next content can be computed from the current content alone, without looking at the micro state: $\pi \circ f = g \circ \pi$ for some content dynamics g . A digital computer's logical level is closed by design. The next content $\pi(f(x))$ of a carrier is what its readers take from it next.
- *Noise margin.* A readout has a noise margin on a set of states when each state in it has a neighbourhood on which the readout is constant, as any voltage well above a bit's threshold reads as 1.
- *Graded dependence.* Content depends gradedly on a region when arbitrarily small changes to that region can move the next content. Formally, no neighbourhood of the region's micro state leaves the next content unchanged while the other regions are held fixed.
- *Contraction.* The overlap discrepancy *contracts* near a state when one step of f shrinks it by a factor below one throughout a neighbourhood of that state, so perturbations too small to cross any threshold are pulled back toward agreement along with larger ones.
- *Physical enforcement.* Agreement is physically enforced at a state when content depends gradedly on each overlapping region there and the overlap discrepancy contracts nearby. These are two separate requirements: feeling small changes, and pulling disagreement back. P applies them carrier by carrier and window by window: in each content window some carrier reaches across each overlap in each direction, and the coupling restores agreement within the window.

The margin lemma. In words: content read through a margin cannot feel small enough physical changes. Precisely, if the micro dynamics is continuous and carries a state into a margin’s operating set, the next content depends gradedly on no region’s micro state. The proof takes one line: continuity carries a small enough neighbourhood of each region’s micro state into the neighbourhood on which the readout is constant. A finite word of thresholded quantities has a margin away from its thresholds, so the lemma covers contents read from bits away from their thresholds. The lemma is mathematics; why it should matter for unity is argued below.

Noise. A noisy threshold’s chance of flipping can vary smoothly with its input voltage even inside a nominal margin. Its next-content probability can therefore retain a small physical dependence that the exact margin lemma discards. P is read on response probabilities relative to the source’s noise.

The *response* of a content to a region at scale η is the largest change that moving the region by at most η makes to the probability of any outcome of the next content, a total-variation distance. The yardstick for a large response is the response that a shift of one standard deviation produces in a Gaussian carrier, $2\Phi(1/2) - 1 = 0.38$, and call it *the fluctuation*: a response this large moves the content by as much as the content naturally fluctuates. Graded dependence above the noise floor asks the response to reach the fluctuation for every change of the region that is larger than its own noise and smaller than the change that crosses one of its thresholds. When the region moves a carrier linearly under additive noise whose density has a single peak, Gaussian or heavy-tailed, a larger change moves the content at least as much as a smaller one, and the criterion is met exactly when a change of one noise amplitude reaches the fluctuation (Appendix C treats noise whose law changes with the state). No constant is fitted to either substrate.

The noisy margin theorem. The margin lemma has a noisy form, and it is this form that carries the argument. Let each realization of the noise

fix an L -Lipschitz step of the micro dynamics, one that stretches no distance by more than a factor L . Let the content be constant within radius r of the noisy successor, except on a set of realizations of probability at most ε . Then a change of one region smaller than r/L moves the probability of every outcome of the next content by at most ε . For anything computed from bits, ε is the probability that some bit of the noisy successor lands within δ of its threshold, and the bound is attained. We call ε the *margin exception probability*. It counts every near approach to a threshold, including those that still produce the right bit, so it can exceed the chip's error rate and has to be checked on its own. A readout fails the response criterion when two things hold: ε is below the fluctuation, and some change larger than the source's noise still fits inside r/L . That change is then a witness, big enough to count and too small to matter.

The simulation objection. A digital consensus algorithm can pull disagreement between contents back, and a simulation can follow a field's course more precisely than biological noise over any window we observe. The course of the contents alone therefore leaves the two unseparated. P looks one level down, at the physical carriers of those contents: the bits inside a simulation, held within their margins, fail the response test even while the simulated contents agree. Which carriers are tested, and how, is therefore part of the argument. Nudging a graded input to the simulation and nudging the voltage of one of its internal bits are different tests, as the replacement analysis below shows.

Wholeness and its realization. On the wholeness view, experiences are unified when they are parts of a single total experience (Bayne, 2010). Their belonging together is a relation among the contents, not one more item represented alongside them. A physical account of that relation has to say which carriers take part, how they act on one another, and over what time they form a whole. Counting answers that the carriers are sensitive to one another. The

motivation is the springs of the introduction. When several regions describe the same graded variable, a coupling can hold their descriptions together through every variation physics can tell apart: move one, and the others respond, while a restoring pull keeps the shared variable consistent.

Why this criterion and not another. Each part of P can be seen at work by adding the parts one at a time and asking what each rules out. Plain causal dependence lets in almost everything: a one-way relay, a common broadcast, a clerk who reconciles the books once a day, and a digital circuit, since changing an input bit changes what its readers compute. Asking that the dependence run both ways and arrive within the content's own window rules out the relay, the broadcast and the slow clerk: an influence that runs one way, or lands after the content has changed, enforces nothing about that content. The window is set by how long the content lasts, so it bites hardest when the world sets the pace. A nation of citizens relaying a brain's signals by radio (Block, 1978) could run the right program, but put in a brain's place, with a human reaction time at every relay, it would still be computing where the bird was long after the bird had gone. Asking the coupling to contract disagreement rules out circuits that amplify it. What survives is fast, reciprocal, restoring dependence, and a digital consensus algorithm on a chip has all three, since its logic settles in nanoseconds. Only the last requirement, sensitivity across the whole range between noise and threshold, separates it from cortex. The margin condition therefore does one job, and its hardest test is a fast, accurate digital replacement (Appendix E).

Counting: one whole, measured at the noise floor. Two carriers are *directly coupled* when every admissible change of either, larger than its own noise and smaller than its threshold, moves the other's next content by at least the fluctuation. Changes are applied to the coordinates that carry content, in both directions where both are possible, and a carrier must admit at least one such change for the test to apply. A *constitutive component* is a

set of carriers joined by chains of direct couplings.

Three things recommend this criterion. First, it intervenes on a variable the system itself reads, so the verdict belongs to the system and survives a change of the observer’s decoder. Second, the noise and the threshold mark out the range over which a carrier can vary without changing category, and we propose that graded contents hold one another in place when each responds to the other across that whole range. Third, chains let an intermediate graded reader join two regions, and the window ties the relation to how long a content lasts. The criterion’s strength is that one intervention protocol applies to every substrate. Its commitment about experience is that unity requires this kind of constraint even when a coarser causal organization produces the same reports.

Records in the coupling graph. A *record* is a content held by a margin. Records still take part in ordinary cause and effect, through changes of their symbols. Under the noisy-margin hypotheses above, counting treats them in three ways.

- A region whose content is held by a margin, with an exception probability below the fluctuation, is coupled to no region that admits a change above its noise inside the margin, whatever the regions it reads carry. It shares a component only with itself.
- For any set of such records, two regions share a component exactly when a chain of couplings avoiding every record joins them. A rule that reads records and writes records adds more isolated vertices, not links.
- When a region’s next content depends on another only through such a record’s symbol, a change inside the margin moves it by at most the record’s exception probability. Writing a correction back through that symbol keeps the same bound.

Graded readers join. A region whose content moves with two regions above their noise, and moves them in turn, joins their components even when they share no direct coupling. Under linear dynamics with uniform noise such a reader joins two regions that a record reading the same two leaves apart (Appendix K). A register holding the bird’s location as a number is a record; a cell that integrates spike times from the visual and parietal areas and projects back to both is a candidate graded reader. It joins them when the measured responses meet the criterion in each direction. A workspace built from reliable records can still make contents jointly accessible; on P, whether they form one physical whole is settled by the coupling graph and the restoring dynamics.

A precedent: clusters in a gas. Physics already counts systems by comparing an interaction with the parts’ own fluctuations. In the theory of imperfect gases, two molecules belong to one physical cluster when their attraction outweighs the kinetic energy of their relative thermal motion, and a cluster is a set of molecules joined by chains of such bonds (Hill, 1955; Sator, 2003). Molecules whose attraction is weaker than their thermal motion still interact, and that interaction changes the pressure of the gas, yet it joins them into no cluster. The precedent is the method: compare a coupling with the parts’ own fluctuations, and join parts through chains. The gas criterion counts bound clusters of molecules; counting applies the same method to carriers of content, and its bearing on experience rests on the argument above.

From coupling to P. When overlap is symmetric, graded dependence across every overlap in both directions is equivalent to *direct mutual coupling* across every overlap. Membership in one component then follows through chains, though a chain can join two regions that have no direct link. For regions with many carriers, the condition asks only that some carrier reach each overlapping region, in each direction, within every content window. The

two directions may use different carriers and moments, which is weaker than asking every carrier to pass, as a formal example shows. P then adds that the coupling must shrink disagreement in time. The connectivity result covers the sensitivity half, and restoration is tested separately.

The chip, counted. A logic node of 1 fF at room temperature has thermal noise of 2 mV (Johnson, 1928; Nyquist, 1928), and the textbook inverter’s margin is 506 times that. A change of one noise amplitude flips the bit with probability below 10^{-100} (Kish, 2002). With margins maintained at the readouts, the response test isolates each bit in the constitutive graph. The bits still act on one another through their logical updates, which run on the chip’s interacting voltages; what they lack is a response to small changes in one another (Appendix D).

Where the line falls. The same test is applied inside each substrate. A membrane’s spike time can respond gradedly near threshold, while a count read through a maintained margin is a record. The membrane calculations below find how often a membrane is in a sensitive state, and an analog crossbar’s column currents are another candidate carrier. Whether any of these contributes to P also depends on what downstream cells or arrays read, and on whether the shared content is restored (Section 5).

The membrane, counted. A spike time passes the test at the moment the membrane rises through threshold, where one unitary cortical input shifts the spike by 1.5 times its jitter (Appendix C). To see how often a membrane is in such a state, the protocol applied to the logic node is applied to a leaky integrate-and-fire membrane driven by white noise, over every state it occupies, with time constants of 5 to 20 ms (Destexhe & Paré, 1999; Destexhe et al., 2003) and rates of 0.5 to 8 Hz (Hromádka et al., 2008). It sits on average 2.0 to 3.5 noise amplitudes below threshold and passes at 0.013 to 2.4% of the states it occupies (Appendix F). A bit restored to a rail

passes at none. No single bit can pass when the noise of the step that reads it includes one amplitude of the change: asked in both directions, its response under Gaussian noise is at most $\Phi(1) - 1/2 = 0.34$, the *binary ceiling*, below the fluctuation.

Rarely at an instant, often within a window. A stationary membrane enters a passing state within 400 ms, the integration time that precedes a conscious percept, with probability 0.085 to 0.997, even when the region it depends on supplies only 1/16 of its noise variance. At least one of 10 independent carriers then reaches the target region within such a window with probability at least 0.59, while at a single instant even 1000 carriers does so with probability as low as 0.025. That is why P is stated per window. The independent-carrier sweep also covers shorter windows. The 20 ms threshold for perceiving the order of two events (Hirsh & Sherrick, 1961) supplies a comparison scale: at least one of 1000 independent carriers reaches the other region more often than not from a window of 3 ms upward, and at 20 ms with probability at least 0.98.

Recorded membranes. Whole-cell recordings from 86 excitatory cells of the barrel cortex of awake mice in quiet wakefulness (Kiritani et al., 2023) constrain a membrane model’s noise, time constant and slow drive. Applying the response protocol to that fitted model tests the declared ranges (Appendix F). The median cell sits 3.7 noise amplitudes below its threshold, farther than the declared 2.0 to 3.5, and its fitted model passes at 0.15% of sampled states, inside the declared range. That model predicts 0.47 times the median cell’s recorded firing rate: recorded membranes fire more readily than the Gaussian model from which their passing states are read. At a noise share of 1/16 it enters a passing state within 400 ms with probability 0.15, and at least one of 10 independent model carriers drawn from these cells reaches the target within that window with probability at least 0.91.

Readers, shared input and switching nodes. Three variations leave the verdict where it is (Appendix F). A reader that registers only whether the membrane fired within the window finds it passing at 0.013 to 2.4% of its states; if every cortical reader pooled spikes only over windows longer than a content’s time scale, P would deny unity to cortex, and that would count against P. With shared input (Poulet & Petersen, 2008) up to 0.75 of the noise variance, at least one of 1000 model carriers supplies a directed reach within 400 ms with probability at least 0.98. And a chip’s nodes, which pass through their thresholds many times in every window, pass at the voltage as a membrane does, since a change of one noise amplitude moves a crossing by one jitter. The voltage is not what the chip reads, though. Timing closure keeps every transition outside the setup-and-hold window of the latch that reads the node (Ginosar, 2011), so the latch reads the restored bit, which passes at no state and in no window. The separation holds at every yardstick from 0.01 to 0.6 (Figure 2). What separates the membrane from the restored bit is the distance from threshold at which each is held when it is read: a few noise amplitudes, against 506.

Categories. Categorical contents, such as the dominant percept in binocular rivalry, have margins in every substrate, so P is stated on the graded variables from which categories are read, such as the relative strength of competing percepts. A graded variable stored in a reliable word of bits is read through margins again.

Gradual replacement. Replacing neurons one by one with units that behave exactly as they do tests any substrate condition (Chalmers, 1995). The hardest case is a recurrent controller whose resolution is finer than biological noise, with enough bandwidth and the natural delay. Its output can respond gradedly, above noise, to a cortical input, even while every bit inside it keeps a wide voltage margin. P therefore looks at the carriers of content and their readers at each interface, and its digital exclusion applies only where mar-

gins are maintained there. Appendix E works this through, including what preserved reports would mean.

What the premise rests on. The case for P is that unified graded contents must hold one another in place, physically, across their whole range of variation, and be pulled back into agreement before they change. The formal results turn that into a precise consequence for readouts held within margins. The membrane computations, including fits to recorded potentials, show that cortex has candidate sensitive states. The bridge tests the proposal together with a calibrated readout of unity, against a model that explains integration by bandwidth, delay and nonlinear transmission (Section 6). Other accounts of unity, such as joint access (Bayne, 2010), make their own predictions once each is given an implementation and a readout model.

4 When and how far a coupling enforces agreement

P asks two things of the coupling: it must pull the disagreement of unified contents back before the content changes, and it must reach every pair of regions whose contents are unified. The first gives the prediction that the restoration test of Section 6 checks; the second says over what distances that test should find restoration.

Agreement in time. A correction that arrives after the content has changed enforces agreement on a content already replaced, so enforced agreement has a deadline. Two things limit how soon it arrives. It arrives no sooner than the coupling’s causal cone allows: a site can track a variable at another only after enough rounds of the coupling to place the second site in its causal past. And it arrives at the coupling’s relaxation rate: a coupling that shrinks the overlap discrepancy by a factor $\kappa < 1$ per step brings a discrepancy of d_0

within ε after $\log(d_0/\varepsilon)/\log(1/\kappa)$ steps, and one that shrinks it by no more than ρ per step needs at least $\log(d_0/\varepsilon)/\log(1/\rho)$. In cortex, signal travel typically takes 18 to 31 ms of the 400 ms that precede a conscious percept, so the deadline falls on the relaxation time (Appendix B). P therefore predicts a cortical relaxation time of at most 160 to 166 ms across the cortex, as signals take the slower or the faster fibres, and 171 ms within a posterior zone. A restoration slower than these counts against P.

Reach. The regions to be joined lie up to 15 cm apart if a scene’s contents span the cortex, and about 5 cm apart if they are confined to a posterior zone. On a sheet of coupled oscillators sharing one natural frequency, in the harmonic approximation, the variance of the phase difference between two sites is the noise strength times their *effective resistance*, the resistance a meter would read if each coupling were a wire of the coupling’s conductance. Under coupling of bounded range the resistance grows at least logarithmically with distance, the Gaussian form of the absence of true long-range order in two dimensions (Mermin & Wagner, 1966), while a kernel whose tail falls off more slowly than r^{-4} bounds it at every distance (Chung & Fuchs, 1951; Kunz & Pfister, 1976). A Lipschitz encoder carries phase agreement to content agreement: at noise strength D the root-mean-square content discrepancy is at most $L\sqrt{D R_{\text{eff}}}$ (Appendix A). In cortex the long-range coupling is synaptic, since the field of a local source falls below detection within millimetres (Appendix I).

Local unity and switching. The regions whose contents are unified may form one patch of cortex, or several patches joined through a hub. Counting then looks at the carriers entering and leaving the hub. A hub that passes only records drops out of the graph and leaves the patches disconnected, one way to realize the switch model of the split brain (Bayne, 2010). A hub whose complete input–output response passes the graded test can join the patches, even if it attends to them in turn, provided its readers’ timing and

memory still let disagreement be pulled back within a content window.

The thalamus is the natural candidate hub. Its relay cells fire in two modes (Sherman, 2001). In the tonic mode the output follows the input nearly linearly (Lesica & Stanley, 2004). In the burst mode the cell answers with an all-or-none burst, typically when excitation follows a prolonged inhibition (Lesica & Stanley, 2004), and it can burst again only after a renewed hyperpolarization (Sherman, 2001). The protocol of Section 3, applied to a model relay cell with both modes (Smith et al., 2000) at matched rates, orders them: within 400 ms a tonic cell enters a passing state at least 2.3 times as often as a bursting one, in every declared regime (Appendix F). Tonic mode therefore offers more chances for a passing route, a prediction to test on relay cells recorded in each mode. P predicts an effect on unity only where that difference decides whether the whole route meets the window and restoration requirements. The same protocol applies to a router between analog patches, whose representation and timing decide which graded changes reach its downstream readers. A bridge across a putative hub tests that whole route.

5 Current artificial systems

A clocked digital chip stores records protected by noise margins and updates them through causal connections between gates. When its readouts stay within the noisy-margin hypotheses, small voltage changes produce responses below the yardstick. A transformer running on such hardware inherits this for its internal logical contents, even when its algorithm pulls disagreements together. A chip tracking the bird holds each area’s estimate as a word of bits, and the two words agree when a step of the program copies or averages one into the other; between steps, neither moves with the voltages beneath the other. Hardware whose contents are graded physical quantities coupled without barriers, such as analog, in-memory or neuromorphic substrates, can

satisfy P. Satisfying it means that such a substrate’s contents, the column currents of an analog crossbar for instance, are held together physically; it does not determine whether any of them is experienced, or whether any is like a human percept.

Symbols and physical contents. The line P draws is between symbolic and physical content, and the line between digital and analog hardware is its most common instance. A *symbol*, in the sense the margin lemma makes precise, is a content read through a noise margin: a physical state whose identity survives every small enough change of the state that carries it, so two symbols agree only when a rule makes them agree. A *physical content* is a graded quantity that the system’s own dynamics reads, so its agreement with the contents it overlaps is produced by the coupling between them. A bit and a spike count latched at a clock edge are symbols; a spike time, a membrane potential and the phase of a coupled oscillator are physical contents.

Organisation, not material. The criterion concerns how a substrate is organised, not what it is made of. An analog crossbar whose column currents settle through shared conductances, or an array of coupled electronic or optical oscillators, couples its contents as a cortical sheet does. A neuromorphic chip passes if its coupled membrane potentials do. A latched count is judged at its reader, with that reader’s noise, margin and integration window, and a neural circuit whose readouts kept the same margins would be excluded in the same way. In analog in-memory accelerators, a layer’s products are currents summed on the lines of a memory array and then digitized: in a 64-core phase-change chip each row’s current is converted into integers, and the traffic between cores runs on those integers (Le Gallo et al., 2023). The words passed between cores are tested at their logical readouts; whether the channel as a whole couples the analog variables of two arrays is tested at those arrays’ readers, as for the cortical bridge.

Why spikes are not bits. What makes all-or-none spikes graded carriers is their timing. When a membrane rises through threshold, the moment it crosses moves continuously with any small input, and receiving cells integrate spike times as they arrive. A clock alone does not separate the substrates: a spike time read through a clock still passes near threshold. What a digital circuit adds is restoration. Each gate drives its output back to a rail, so every transition a latch reads starts and ends far from any threshold. Clockless logic fails in the same way: its transition times move with every voltage, but its values are read from completed codes held at rails (Appendix G).

Leakage and the margin's width. Supply noise, crosstalk and temperature do move the voltages beneath digital readouts. While the theorem's hypotheses hold, their effect on the next logical content is capped by the margin exception probability. Each gate pulls its output back to a rail, and the logic network coordinates values across rails; this makes no bit respond to small changes in another. The width of the margin decides which physical disturbances the bound covers, and the precision of a represented number is a separate matter from the voltage margin of each bit (Appendix G).

Attractor basins. Cortex has margins of its own. An attractor network's basin is a dynamical margin: which attractor the network settles in survives every small enough change. The basin gives the attractor's label a margin, but it leaves graded the state that carries the label: the position along a continuous attractor, such as a bump of activity that encodes a direction, and the membrane potentials and spike times within the basin, which downstream cells integrate. A cortical circuit whose contents were read only as attractor labels, far from the basins' boundaries, would fall under the margin lemma as a chip does, and the weak nudge of Section 6 tests which case cortex is in.

6 Tests

The readout assumption. A bridge changes a physical mechanism, and inferring anything about experience from it also needs a model of observation. *Assumption R:* within the tested interventions, a subject’s calibrated reports of one simultaneous field spanning the two regions remain evidence about experienced unity. Calibration separates those reports from reports of local visibility, the ability to report from each region, attention, and success on each component task. The criterion is fixed in advance and applied equally to natural coupling, artificial bridges and gradual replacement.

Under R, if a bridge certified to remove P’s coupling leaves reports of unity intact, P and R are challenged together, and the experiment commits to reading that outcome as a challenge. The same reading applies to gradual replacement. In animals, matching across two fields measures the integration of information and tests the mechanism; a human preparation adds reports of unity. Once the companion has validated it, decoded agreement serves as a neural correlate, with its decoder held fixed across interventions. The limits of R are stated in Section 8.

Separating coupling from information. In every manipulation of cortex available so far, graded coupling and information move together, and stimulating the regions from outside does not separate them. Coherence between cortical regions arises largely from communication between them (Schneider et al., 2021), and where in-phase transcranial stimulation of two regions raised their synchrony, it raised directed information flow with it (Reinhart & Nguyen, 2019). The coupling has to be supplied instead.

The bridge. Interrupt the natural connection between two regions, and reconnect them by stimulating each in proportion to the activity recorded at the other (Figure 1). The reconnection runs through one of two loops.

- The *analog loop* is an analog filter and amplifier driving the stimulator,

analog end to end, so its output moves with every graded change in its input.

- The *quantized loop* is the same filter and amplifier with a quantizer between them, so its output is read from bits and has a margin.

Both loops share the output stage and stimulation path. The design is factorial: each bandwidth is run with and without the quantizer. Within that family one pair is chosen to control information, as described next, and further blocks sweep the quantizer’s step at fixed bandwidth and delay.

Matching the information. In the information-controlled pair, the analog loop carries no more estimated transfer entropy or task information than the quantized loop; its bandwidth is lowered to achieve this while it still settles within the content window (Appendix H). Whether each loop passes P is a separate question, answered on the full circuit: each direction is tested at the actual downstream reader, including that reader’s memory and integration time. Transfer entropy counts everything one region’s activity tells about the other’s, and the loops could match it and still differ in how much of the task gets through. So the task variable is also decoded from the activity each loop evokes in the region it drives, and the loops count as matched when the analog loop carries no more information than the quantized one, in total or about the task. Both loops were simulated on a model pair of regions, with a linear and a nearest-neighbour estimator of each quantity. At a step of 10 noise amplitudes, a cutoff between 17 and 95 Hz matches the linear estimate of transfer entropy in every simulated regime whose task variable spans at least half a step. At that cutoff the analog loop carries less task information than the quantized one by either estimator, and less transfer entropy by the nearest-neighbour one. Matching these two summaries is a control and leaves the outcome open: the dynamical alternative below works from the full delivered signal and can still predict different outcomes for matched loops.

A competing mechanism. The alternative is a recurrent observer. Each region’s estimate combines its own local evidence with the signal the bridge delivers, updated with a finite delay, a fitted gain and noise, and a decision rule reads the difference between the estimates (Appendix H). The filter and quantizer enter only through their measured effect on the signal. The observer’s parameters are fitted on pilot blocks and then frozen, and it is scored on held-out bandwidths, quantizer resolutions and perturbations, predicting both choices and recovery from disagreement. It can favour either arm through timing and nonlinearity alone, and gives no special role to sensitivity inside a margin.

Certifying the contrast. P asks for a carrier response somewhere within each content window, so each loop is classified by the probability of visiting a passing state in each direction, together with the measured restoration of disagreement. A loop can spend little time in passing states and still visit them often. Every candidate route counts: threshold crossings, joint words, repeated samples and residual biological paths. To exclude a loop, the response bound must cover the complete map to the next readout, memory included; to include one, the measured carriers must also restore agreement (Appendix H). Both arms use the same noise scale, interventions and content window.

Predictions and outcomes. Take blocks in which the analog bridge restores the calibrated unity readout. P with R predicts that a quantized bridge certified to lack the required coupling will lose that readout. The recurrent observer predicts the readout from the transmitted signals through its frozen decision rule. The decisive blocks are those in which the two predictions diverge while local visibility and report ability stay intact. Preserved unity reports there challenge P with R. An analog advantage there favours P only if the observer fails to account for it within its preregistered uncertainty. If both bridges pass the coupling test, equal performance is what P expects. If

the analog bridge fails to restore integration, the positive control has failed and the comparison is uninformative. In animals, accuracy constrains the observer’s predictions and the perturbation measurements test P’s physical conditions; the experiential comparison rests on R’s calibration in humans.

A first preparation. Closed loops that trigger stimulation at one cortical site from spikes recorded at another already run for days in behaving animals (Jackson et al., 2006; Guggenmos et al., 2013); the analog arm replaces the trigger with a continuous drive. A first preparation places chronic arrays in two areas that represent a shared variable, such as the visual and parietal areas that represent the bird’s location, interrupts the natural connection between them, and alternates bandwidth and quantization conditions across blocks in the same animal. Directed perturbations check for residual routes before the interruption is treated as a disconnection. Transfer entropy (Schreiber, 2000) is estimated from each block’s recordings, and the step and cutoff are set on pilot blocks. The primary animal readout is behavioural integration: a task requiring the values the two areas represent to be combined, such as judging whether stimuli in their two fields match. Blocks with the connection interrupted and neither loop running come first, as a positive control: the comparison is informative only if the analog loop restores some of the integration that interruption removes. The companion’s excess decoded agreement is the neural readout. At the declared effect sizes the behavioural comparison is the weaker, needing 1248 trials per loop for a pairwise comparison; the factorial design and the equivalence margins for preserved reports need a power calculation on pilot data. The loops record and stimulate in separate frequency bands, so the recordings stay readable under a continuous drive (Appendix H).

A weak nudge should spread. Nudge a content-bearing variable in one region, in either direction, by anything from the source’s noise amplitude up to its threshold, and measure the next content read in an overlapping

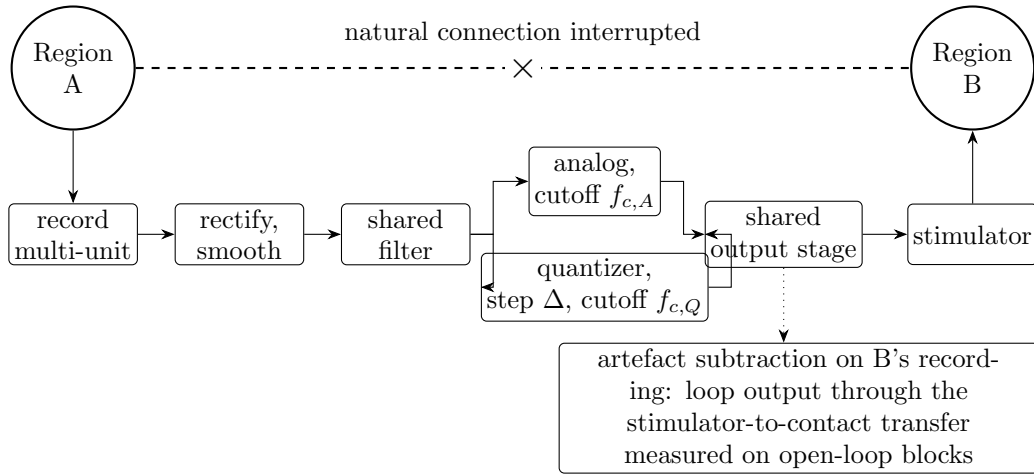


Figure 1: The bridge, one direction of two. Multi-unit activity recorded in region A is rectified and smoothed, passed through a filter both loops share, and then either kept analog or quantized, before an output stage both loops share drives the stimulator in region B. The arm cutoffs $f_{c,A}$ and $f_{c,Q}$ are varied in both arms. The displayed pair uses the quantizer’s step and the analog cutoff to control estimated transfer entropy and task information; the full comparison also tests both arms at common bandwidths. What leaks from the stimulator into a recording contact is removed by subtracting the loop’s own output, passed through the measured transfer. The loop from B to A is the same.

region. P predicts that within the content window some carrier responds by at least the declared yardstick, and that the shared variable moves back toward agreement. Transcranial alternating current and weak intracortical fields are candidate interventions. The decoded variable is graded, read by the region’s own targets, and calibrated against the source’s noise. A response that widens the disagreement challenges the restoring half of P; a valid upper bound below the yardstick throughout the window challenges its sensitivity half. The rival observer is tested on the same intervention time courses.

Agreement should be restored. Perturb one region and measure how fast its disagreement with its partners returns to the declared tolerance. P

predicts restoration within the relaxation time of Section 4 for the admissible perturbations, with proportional restoring response in the linear regime. On an artificial system two kinds of perturbation are possible. Patching a represented activation changes a logical value and tests whether the algorithm restores agreement. Nudging an internal physical carrier within its margin tests the response bound. They act on different variables, as the replacement analysis shows.

The field’s causal cone. The brain’s electric field reaches the tissue with negligible propagation delay on neural time scales (Gratiy et al., 2017), while synaptic influence across the cortex takes at least 8 ms. Re-agreement faster than the synaptic route allows would reveal a non-synaptic part of the coupling. The field of a local source falls below detection within millimetres, so the cone is tested at millimetre distances, and at centimetres P predicts re-agreement through synaptic coupling, which is graded as well (Appendix I).

Split brain. Callosal section, read as divided perception with undivided consciousness (Pinto et al., 2017), contested (de Haan et al., 2020), or as consciousness alternating between the hemispheres (Bayne, 2010), removes graded coupling and information together, and the remaining subcortical routes carry both at once: the confound the bridge is built to break.

7 Related work

Appendix J relates P to accounts that tie consciousness to biology, to the analog–digital distinction and to field theories.

Indicator properties. Butlin et al. (2023) assess current AI systems against indicator properties of consciousness, each stated computationally, so every indicator sits at the grain of an algorithm. P adds one indicator stated at

the grain of graded dynamics, a condition on how the substrate realises the others, so a system can have every computational indicator and fail P, as a faithful simulation of cortex does.

Computational closure. On the account of Rosas et al. (2024), a macroscopic level that is informationally, causally and computationally closed marks emergent software in a physical system. That closure is the property Section 3 finds at the content level of a digital system, and P reads it the other way round: a level closed off from the graded state beneath it is one whose agreement is computed, not enforced.

Designed to suppress its physics. Kleiner & Ludwig (2024) argue that if a system’s physical dynamics is relevant to its consciousness, processors designed to suppress deviations from their computational dynamics cannot be conscious. The margin lemma states that suppression formally, and this paper adds a scale fixed by the parts’ own noise, which counts systems instead of asking whether physics is relevant at all, a contraction condition, and a test in cortex.

Integrated information. Integrated information theory also denies experience to digital simulations of a brain (Tononi & Koch, 2015; Findlay et al., 2024), through the causal structure of the hardware at the grain of its logic gates. P proposes sensitivity and restoration of graded content-bearing carriers as its constitutive condition, evaluated through the same response protocol on model membranes and logic nodes.

What physical tests can tell us. Love (2026) argues that third-person science cannot resolve why physical processes are experienced or settle the attribution of experience to another system, while its findings can inform practical judgments about that attribution. This paper works within that distinction: P is a proposed necessary condition for the unity of contents

already taken to be conscious, and the bridge tests its physical mechanism together with an assumption about reports. Neither the response bounds nor their experimental tests solve the hard problem. A Russellian-monist reading, on which the intrinsic nature underlying physical dynamics matters to experience (Alter & Pereboom, 2023), is compatible with this use of measured dynamics to inform judgments about other systems; P does not identify that intrinsic nature.

8 Limitations

- *The premise.* P is a premise and is not derived from the dynamics; the cortical tests bear on whether it holds. It is necessary for unity and not sufficient for consciousness: coupled pendulums satisfy it. It constrains how a system’s conscious contents are held together and is silent on which contents are conscious and what they are like. An analog cross-bar that passes P holds its column currents together physically, which makes none of them an experience, still less one like a human percept.
- *Wholeness and counting.* Wholeness does not entail counting or contraction. Counting is a hypothesis about how unity is realized, stricter than ordinary causal dependence and stricter than failure to split into independent parts. The logical states of a digital network depend causally on one another, and a field may have a closed effective description. Our reasons for asking for sensitivity across the whole admissible range motivate P but do not prove that experience requires it. The gas analogy supports comparing coupling with fluctuations, not identifying physical clusters with subjects. Rejecting counting leaves the response bounds standing and removes their reading in terms of experience.
- *Grain and functionalism.* A functional account that includes the same response and restoration criterion agrees with P. Accounts built on tem-

poral organization, nonlinear transmission or joint access can still reject its claim about what unity consists in. Matching two information summaries leaves those organizations different, so a successful alternative model remains a competitor, even when it tells the loops apart.

- *The hardware counts.* The digital exclusion needs the margin, Lipschitz and admissible-change hypotheses to hold at the actual next readout. The margin exception probability counts near approaches as well as errors, so it is generally larger than a logical error rate. The single-bit ceiling holds for the stated symmetric single-peaked noise laws; a Student law with three degrees of freedom can break it. A word of bits read jointly, or a sequence integrated over time, needs its own response law. At a cortical interface, a high-precision converter can pass on perturbations above biological noise even though the processor's internal bits keep large margins. The thermal and switching examples declare their capacitance, clock, activity, transition time and synchronizer.
- *Readouts and replacement.* R is an additional assumption linking calibrated reports to unity under the interventions tested. Behaviour can integrate information without the experience at issue here, so an animal matching task does not establish experiential unity, and a human report of unity needs visibility, attention and report ability controlled. If reports of unity survive a certified failure of P, P and R are challenged together; giving up R would also give up the behavioural inference in gradual replacement. The proposed experiment cannot settle P independently of every readout assumption.
- *The cortical estimates.* The single-input estimate combines in vitro measurements of one connection type and one noise recording, through a linear threshold crossing. The occupancy computation models one integrate-and-fire membrane under white noise, with a threshold without dynamics of its own, and its time constants, rates, windows, noise

shares and carrier counts are declared ranges, not fits. The recorded check covers one area of one species in one dataset without simultaneous recordings, and reads its passing states off a Gaussian model held at the slow part of the potential, which predicts 0.47 times the median cell’s firing rate. Correlation is one shared component at declared shares, compared with the correlations of neighbouring cells, and omits the unshared input that drives the single spikes of awake cortex (Poulet & Petersen, 2008). The carrier counts stay far below the anatomical ones. The per-window bound is proved for independent carriers and, without independence, as the variance of the number that pass over the square of its mean, a ratio that correlation keeps above a floor. The sweep of the window’s length assumes independent carriers, and shared input lowers the link most in the shortest windows. The relay cell is a reduced model with no reticular feedback. The passing-state estimates measure candidate sensitivity, not reciprocal contraction of a network’s decoded contents. The time windows are declared perceptual scales; the temporal-order threshold supplies a benchmark rather than a universal lower bound on content change.

- *Reach.* Phase variance equals effective resistance in the harmonic approximation at identical natural frequencies. The logarithmic lower bound is proved for kernels of bounded range, and the bound under a tail slower than r^{-4} for a sheet that supplies the site counts and couplings of a dyadic chain of discs. Unbounded resistance under a kernel of finite second moment is the classical recurrence criterion on the lattice; logarithmic growth is established numerically. Above the order–disorder crossover the harmonic identity fails. One oscillator population per content patch is a modelling choice about cortex (Appendix A).
- *Which variables are contents.* Which variables a system’s own dynam-

ics reads is settled system by system from its circuitry, and the margin lemma takes it as given. A cortex whose unified contents were attractor labels alone would fail P.

- *Artificial systems.* Confirming the predictions for artificial systems checks that hardware runs inside its margins and leaves P untested; the tests of P are the cortical ones.
- *Agreement in time.* Its edges are proved for a given contraction factor, and identifying that factor with the coupling's spectral gap is linear response. The relaxation time of cortical coupling is unmeasured, and agreement is read as a declared reduction factor of disagreement.
- *The field.* A long-range tail improves agreement only in proportion to its conductance against the short-range part. The field's reach rests on an estimate that places its peak at a declared distance, and its contribution across centimetres is a quantity to be measured.
- *The bridge's reach.* The saved simulations show that information can be controlled in a one-directional transmission model with declared noise and filters. They do not establish that the full reciprocal preparation falls on opposite sides of P, and they do not fit the rival observer. The model has no spikes, stimulation artefact or plasticity. The fraction of time a loop fails at an instant does not certify that it fails over a whole window, and bounds on single bits do not certify a reader of a joint word or of repeated samples. Temporal response, alternative anatomical routes and restoration need measuring. The factorial comparison and resolution sweep are proposals, with power and equivalence margins to be set on pilot data.
- *Interpreting outcomes.* An advantage for the analog arm is not enough evidence for P on its own, since bandwidth, delay and nonlinear transmission may explain it; the comparison needs predictions that diverge

from the fitted alternative. A null result challenges P with R only when a bridge certified to lack the required coupling preserves the calibrated unity readout, with enough power to establish equivalence. If both bridges pass, equal outcomes are compatible with P. The positive control requires the analog loop to restore integration, and stimulation can recruit cells synchronously or antidromically. The recorded envelope samples only part of a region’s carriers.

- *Human evidence.* No human bridge preparation here measures reported unity, and the readout and the coupling classification still have to be validated in humans. A hypothesis that ties unity to a particular tissue would be a different proposal from P’s organizational criterion, and the electronic bridge leaves it untested.

Data and code availability

The Lean development, simulation scripts, tests and saved numerical summaries are maintained in the GitHub repository `bobaseb/coupling_gated_cortical_coherence`. The formalization uses Lean 4.34.0-rc2 and Mathlib commit `5ce203ecd53f8c3e47ee5530d7c86b94a55f0183`. Publication simulation values are generated from saved summaries without rerunning production sweeps. The whole-cell recordings are those of Kiritani et al. (2023), deposited on Zenodo under CC BY 4.0 ([doi:10.5281/zenodo.7833080](https://doi.org/10.5281/zenodo.7833080)); the repository fetches and verifies them, contains the analysis, and does not redistribute the recordings.

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Use of AI tools

AI assistance to this work was substantial and is not separable by task. Several coding assistants and chat interfaces were used in an interleaved way throughout: Google’s Antigravity CLI (Gemini 3.1 Pro, Claude 4.6 Opus, Claude 4.6 Sonnet), OpenAI’s Codex CLI (GPT-6 Astra and Sol; GPT-5.6 Terra, Luna and Sol), Claude Code (Claude Opus 5 and Opus 5.5), Gemini Pro through its web interface, and Anthropic’s Fable and DeepSeek v4 Flash through OpenRouter, accessed with the Hermes CLI. Between them they contributed to the Lean development, the simulation and analysis code, and the drafting and revision of this paper. Because the tools were used in combination and in alternation, no section, proof, figure or result is attributable to any one of them, and none is claimed to be. The author is responsible for all content, including everything a tool produced. Machine checking establishes the stated Lean results under their hypotheses; it does not validate biological interpretations or replace scientific review of generated code and prose.

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A Reach: resistance, encoders and simulations

The logarithmic growth is a lower bound for every kernel of bounded range. Group the sites into shells around a site a , so that the coupling joins only sites in the same or adjacent shells. The shells then act as conductors in series, and the effective conductance from a to any site beyond shell n is at most $(\sum_{k < n} 1/C_k)^{-1}$, where C_k is the coupling crossing from shell k to shell $k+1$. On a planar sheet the perimeter of shell k grows linearly in k , so $C_k \leq c(k+1)$ for a constant c set by the kernel’s strength and width, and the effective resistance is at least $\log(n+1)/c$. Widening the kernel enlarges its shells and raises c , which slows the growth and leaves it unbounded. Distance-independent agreement on a sheet therefore needs coupling of unbounded range.

A tail slower than r^{-4} bounds the resistance at every distance. Run a chain of site sets out from a through discs of radius 2^j and back in to b , the two widest discs spanning the separation. The phase difference across a and b telescopes through the discs’ mean phases, and the drop between the means of two consecutive discs is bounded by the coupling energy between them, shared among the pairs of their sites. A disc of radius 2^j on a planar

sheet holds at least $\alpha 4^j$ sites, and a kernel $K(r) \gtrsim r^{-(2+\sigma)}$ couples two sites of consecutive discs by at least $\beta 2^{-(2+\sigma)j}$. The number of pairs grows as 16^j and their coupling falls as $2^{-(2+\sigma)j}$, so the drops shrink geometrically with ratio $q = 2^{-(2-\sigma)/2}$, which is below one for $\sigma < 2$, and the effective conductance is at least $(1 - q)^2 \alpha^2 \beta / 8$ however far apart a and b lie.

Both results match classical potential theory. Imagine a random walker on the lattice that jumps between sites with probabilities proportional to the kernel, which is the same at every site. The resistance from a site to infinity is infinite exactly when that walker is sure to return to its start, that is, when the walk is recurrent (Doyle & Snell, 1984). Shorting every site outside a disc around a can only lower the resistance from a to a site b beyond the disc, so under recurrence that resistance grows without bound as b recedes. A symmetric kernel of finite second moment, which includes every tail steeper than r^{-4} , gives a recurrent walk in two dimensions (Chung & Fuchs, 1951). The bound above is the matching statement for tails slower than r^{-4} , whose walks escape. The same threshold holds beyond the harmonic approximation: the plane rotator in two dimensions with coupling falling as $r^{-\alpha}$, $2 < \alpha < 4$, has true long-range order at low noise (Kunz & Pfister, 1976), which short-range coupling forbids. With short-range coupling the plane-rotator sheet has a Kosterlitz–Thouless transition (Fröhlich & Spencer, 1981), an order–disorder crossover that the simulations below locate. When natural frequencies differ from site to site, short-range coupling fares worse still: phases then desynchronize in every dimension up to four (Hong et al., 2005).

Two further properties bound the resistance from above. Adding coupling never raises it, and a direct coupling caps it at its reciprocal. Across a finite cortex a wide short-range kernel and a slow tail differ by a slowly growing logarithm, so telling them apart takes a measurement of the kernel’s second moment as well as of its tail.

An encoder that reads the phases of a small neighbourhood relative to the phase at its centre does better than the Lipschitz bound of Section 4.

It reports one content at every site of a travelling plane wave or a winding, whatever the gradient, and for a wave plus a departure from it its discrepancy between two sites is at most its Lipschitz constant times the departure's relative readings there, so it pays for the local roughness of the field and for nothing else. Both bounds read one oscillator population per content patch. Across centimetres, where two territories describing one quantity need not be neighbours on the sheet, that reading is an assumption, the *phase-to-content assumption*: each territory's content is read, by encoders of Lipschitz constant L that are equal where territories overlap, from the phase of one oscillator population assigned to it, so that the phase agreement of the two populations bounds the agreement of the two contents. It is a modelling choice about cortex, since the coupling of a field is spatial while two territories describing one quantity need not be neighbours. Under it the coupling's reach bounds phase disagreement between distant territories directly and content disagreement through it.

Simulated sheets at identical frequencies, of side 64 and 128, with nearest-neighbour, exponential and power-law kernels of tail exponent $\sigma = 1$ and $\sigma = 3$ at a common row sum, follow the harmonic identity in the ordered phase. The measured phase variance is at most 1.15 times $D R_{\text{eff}}$ at $D = 0.1$ and at most 1.26 times at $D = 0.4$, the excess rising with noise as the cosine coupling softens, and a logarithm fits its growth with distance better than a line in 16 of the 16 runs at those noise levels. At $D = 1.2$ the nearest-neighbour sheet disorders: the mean chord discrepancy $\langle 1 - \cos(\theta_a - \theta_b) \rangle$ at the largest separation is 1.00. The exponential and $\sigma = 1$ sheets stay ordered at that noise (0.46 and 0.50), and the $\sigma = 3$ sheet nearly disorders (0.92), so the kernel's width sets the order-disorder crossover as well as the rate of decay. Exact resistance sums at sheet sides 128, 512 and 1024 separate width from tail. The exponential kernel, with 25 times the nearest-neighbour stiffness, gives resistance at half the side of 0.531, 0.549 and 0.558, still growing; the $\sigma = 1$ tail gives 0.580, 0.577 and 0.577, and the $\sigma = 3$ tail, steeper than r^{-4} ,

gives 1.198, 1.448 and 1.572, growing as a short-range kernel does.

B Agreement in time for cortex

The causal cone is stated precisely as follows: a report at one site that tracks a variable at another to within ε , for two values more than 2ε apart, is possible after T rounds of the coupling only if the second site lies in the first site’s causal past for T rounds.

The longest cortico-cortical fibres run about 15 cm (Nunez & Srinivasan, 2014). Callosal axons conduct at median velocities of 4.9 to 8.8 m/s, and the fastest of them at up to 20 m/s (Caminiti et al., 2009). A chemical synapse adds about 0.5 ms (Katz & Miledi, 1965; Sabatini & Regehr, 1996). The earliest a synaptic signal can cross the cortex is therefore 8 ms, along the fastest axons, and typical arrival takes 18 to 31 ms. For the time scale of a content, take the unconscious integration that precedes a conscious percept, which lasts up to 400 ms (Herzog et al., 2016). Count agreement as reached when disagreement has fallen 10-fold, decaying exponentially. Agreement then arrives in time when the coupling’s relaxation time is at most 160 ms with typical arrival along the slower fibres, and 166 ms along the faster. Over a localized extent of 5 cm the earliest arrival is 3 ms, typical arrival along the faster fibres is 6.2 ms, and the relaxation time may reach 171 ms. Signal travel thus uses only a small part of the time available, and whether agreement arrives in time depends on how fast the coupling settles.

C A single input and noise that changes with the state

A neuron in numbers. Suppose an input ΔV , too small to fire the cell on its own, arrives while the membrane potential is rising through threshold at rate $\dot{V}$. It moves the spike by $\Delta V/\dot{V}$. Voltage noise σ_V jitters the spike by

$\sigma_V/\dot{V}$. The shift therefore exceeds the jitter exactly when ΔV exceeds σ_V , whatever the slope, and spike generation adds little noise of its own (Mainen & Sejnowski, 1995). A unitary excitatory potential between cortical pyramidal cells averages 1.3 mV, with a coefficient of variation of 0.52 (Markram et al., 1997), and the membrane noise near threshold is 0.54 mV (Jacobson et al., 2005). Counting the input’s own variability as noise, one such input shifts the spike by 1.5 times its jitter, a response of 0.55. Every input larger than 0.64 mV passes, up to the input that fires the cell.

Noise whose law changes with the state. When the law of the noise changes with the state, as the variance of a spike count does, a larger change still moves the content at least as much as a smaller one on each side of the state, provided the laws have a monotone likelihood ratio, as exponential families in their natural parameter do: such laws cross once along the parameter the region moves. The criterion of Section 3 is then met exactly when a change of one noise amplitude up and one down both reach the fluctuation.

D Counting in detail

The logic node. In the declared Gaussian model a change of one amplitude toward the threshold moves a restored bit by the fluctuation only at half an amplitude from the threshold, 0.099% of the margin’s width. When both directions are admissible, their smaller response stays below the fluctuation at every distance, by the single-bit ceiling. The noisy margin theorem supplies the more general exclusion whenever the exception probability is below the yardstick and an admissible perturbation fits inside the effective margin. A readout that meets those hypotheses at every state in a window fails throughout the window.

The circuit still passes on changes of logical state, and crosstalk and supply noise still move its voltages. Counting isolates a reliable record only

with respect to the stated physical interventions; both kinds of interaction remain in the full description of the chip. A reader that integrates several decisions is judged on its complete next-readout map, as in Appendix H.

Effective descriptions and causal dependence. At the logical level a digital circuit has an update $b' = g(b)$. The swap $b'_1 = b_2$, $b'_2 = b_1$ couples its two bits causally: flip either input and the other output flips. Computational closure keeps this dependence and discards the voltage differences within a logical state. Closure, and ordinary failure to split into independent parts, therefore answer a different question from counting, which asks which physical changes that leave the content intact still reach another carrier above noise. A digital circuit's records are isolated in that sense while their logical values depend on one another.

A field can also have a closed effective description. Its case for P rests on how its selected carriers actually respond and restore. For every substrate, the content variables, their noise, the admissible interventions and the reader's integration time are fixed before the test. These choices give the count a physical meaning, and give a critic the quantities with which to challenge it.

E Gradual replacement

The strongest replacement is a recurrent digital controller that reproduces the relevant cortical dynamics to within biological noise, with enough bandwidth and the natural delay. Its converters resolve changes smaller than the noise on its cortical input, and its output drives the same downstream readers. Replacing neurons one by one with such units preserves functional organization at the grain used to predict behaviour (Chalmers, 1995).

Two interventions need to be kept apart. Changing the cortical input by one of its noise amplitudes can move many quantization levels and change the

controller's output. Changing the voltage of an internal bit by one of its own noise amplitudes, inside its margin, leaves the next logical content almost unchanged. The margin theorem bounds the second under its hypotheses; more numerical precision makes the first finer without narrowing any internal bit's margin.

P claims that the internal realization of content matters too. The motivation is mutual constraint among graded carriers (Section 3), and the demanding consequence is that an accurate logical emulation can reproduce the dynamics of content while lacking that internal relation. Applying this to a hybrid means first identifying which internal variables actually carry the relevant contents. Where the controller only relays signals between cortical carriers, those carriers' responses to one another are measured through the whole channel. A finely quantized channel may pass every admissible cortical change; being digital supplies no exclusion on its own.

The resolution sweep therefore does two jobs. It tests the rival observer's prediction that performance improves as transmission error and delay shrink, and it finds where, if anywhere, the cortical interfaces cross the coupling criterion. If the finest bridge passes P there, equal readouts fit both accounts. If it fails throughout the relevant windows while the unity readout stays intact, P is challenged together with R.

The readout assumption is the same for gradual replacement and for the bridge. Taking unchanged reports as evidence of unchanged unity commits P with R to that challenge. A version of P that lets unity change behind perfectly preserved reports rejects R for those interventions and forfeits this behavioural test. Which package to hold is chosen before the experiment and stays fixed whatever the outcome.

F The occupancy computation

What the membrane is asked. The membrane's next content is the time of its first spike within 10 or 30 ms, and every change of one to 3 noise amplitudes, in either direction, must move that content by the fluctuation. Its time constants span the in vivo high-conductance state and its rates the sparse rates of awake cortex. The passing fractions of Section 3 are for a region that supplies all of the membrane's noise. A bit restored to a rail at the logic node's margin passes at none of its states, and its largest response over them is below 10^{-100} . The binary ceiling stays below the fluctuation under every symmetric noise law with one peak that puts less than twice the fluctuation of its mass within one standard deviation of its centre, logistic and Laplace noise included, as shown at the end of this appendix. A graded content clears the fluctuation by a wide margin: at its best state the membrane responds by at least 0.64 in every regime and at every noise share.

Windows. For independent carriers the chance that none passes within a window falls geometrically with their number: with 100 carriers some carrier passes within 40 ms with probability at least 0.59. Reading the spike time through a clock of 1 or 5 ms leaves the passing fraction unchanged. The value a latch reads links in no window of any length.

Readers that ignore spike times. A reader that registers only whether the membrane fired within the window, and discards every spike time, passes with a next content of 10 ms at the same share of states in every regime as a reader of spike times. At a membrane a few amplitudes below threshold, what a change of one noise amplitude moves is the chance that it fires within the window, and a spike count registers that; a count responds at least as much as this reader, since whether the membrane fired is a function of the count. So P classes a cortical membrane as a graded carrier whether its

readers integrate spike times at the grain of milliseconds, as the evidence of Section 2 indicates, or pool spikes over the content’s window.

Shared input. In awake cortex neighbouring membrane potentials are highly correlated during quiet wakefulness and less so during active behaviour (Poulet & Petersen, 2008). The model gives every membrane a shared component carrying a fraction c of its noise variance, so that two membranes correlate by c and each keeps its own law. At least one of 100 such carriers enters a passing state within 400 ms with probability at least 0.88 for every share up to $c = 0.75$, in every regime.

Switching nodes. Take a 1 GHz clock and nodes that switch in 0.02 to 0.5 of its cycles, each transition lasting 20 ps. If a node’s next content were the time of its crossing, the node would pass on the rising half of each transition, at 0.02 to 0.5% of its states, within the membrane’s range, and within every window of 40 ms except with probability below 10^{-100} . By the carrier condition of Section 2, though, a node’s next content is what its latch reads at the clock edge, the node at its rail. The crossing a latch discards is exactly the content a membrane delivers when its readers integrate spike times as they arrive. The separation holds for count readers too: a reader that registered only whether a node switched within a cycle would take a quantity fixed by the node’s logical inputs, which are restored bits, moved by its noise only through the margin exception probability.

A thalamic relay cell. The integrate-and-fire-or-burst model (Smith et al., 2000), with its published parameters, adds to the membrane a low-threshold calcium current that flows above -60 mV, inactivates there and recovers below it. Noise gives the passive membrane a standard deviation of 2 or 4 mV, the unit of every change. In burst mode a constant current holds the membrane below the calcium threshold, where the current recovers and noise that carries the membrane past the threshold sets off a burst; in

tonic mode it holds the membrane above, the current stays inactivated, and the cell is an integrate-and-fire membrane. In each mode the current is calibrated to a rate of 0.5 or 2 Hz, and the next content is the first spike time within 10 or 30 ms, as for the cortical membrane, with the calcium current's state kept under each change. At an instant the two modes pass at similar shares of their states, 0.16 to 1.5% in tonic mode and 0.3 to 1.6% in burst mode. A bursting cell's passing states lie in and around its bursts, which are rare, while a tonic cell's lie below its spike threshold, where it spends the time between spikes. Within 400 ms a tonic cell enters a passing state with probability 0.28 to 0.84, and a bursting cell with probability 0.07 to 0.34. At matched noise, rate and window the tonic cell's chance is at least 2.3 times the bursting cell's, within 40 ms as well. Probability is propagated exactly on a grid in membrane potential and inactivation, and a run at half every grid step moves no visit probability by more than 8.5%.

Recorded membranes. The recordings (Kiritani et al., 2023) hold 240 cells of layers 2/3 to 5 of the barrel cortex of awake head-restrained mice, with quiet wakefulness, free whisking and touch scored from whisker filming; the excitatory cells are the membranes the model describes. Spikes are detected by the authors' criterion for each cell, a cell's threshold is the median potential at spike onset, and each spike, from onset to repolarisation, counts as a failing state, as the model's refractory states do. The awake potential moves on two time scales: fluctuations with a correlation time of 11 to 17 ms, and a wander over hundreds of milliseconds that carries much of its variance. One time constant fitted to both misses the fast fluctuations that set the next spike, so the potential is split into its running mean over 100 ms and a fast remainder. The noise amplitude and the model's time constant are those at which the model membrane, analysed exactly as the recording is, has the remainder's spread and correlation time. Within a next-content window the membrane is taken to be the model membrane with its mean held at the

running mean’s current value, which supplies the law of the next spike time that a single recorded path does not. Across the 86 cells with at least 10 s of quiet wakefulness, the tenth to ninetieth percentiles run from 2.2 to 5.1 noise amplitudes below threshold, from 0.025 to 1.3% of states passing, with 90% of cells inside the declared range, and from 0.01 to 0.51 for a passing state within 400 ms at a noise share of 1/16; for 31% of cells that chance is below the least declared regime’s, 0.085. During free whisking (73 cells) the median cell sits 4.1 amplitudes below threshold and enters a passing state within the window with probability 0.13. Splitting at 200 ms instead gives a median passing share of 0.12% and a median visit probability of 0.13. For the median cell the model predicts 0.47 times the recorded firing rate, and it comes within a factor of two for 48% of cells: recorded membranes fire more readily than a Gaussian membrane at the same distance, and the passing states are read off that Gaussian description. None of the 689 sweeps timed to the second holds two cells recorded at once, so the share of input that neighbouring cells hold in common stays declared.

The yardstick. The separation of the membrane from the value a latch reads holds at every yardstick from 0.01 to 0.6 (Figure 2), not only at the fluctuation. Across that range the latched logic node passes at none of its states, and the membrane enters a passing state within 400 ms with probability at least 0.036 at the highest yardstick, and at least as often at every lower one. Below $\Phi(1) - 1/2$ one bit can reach the yardstick, so there a synchronizer’s first stage can pass, at no more than 0.2% of its samples. The formal results take the yardstick as a parameter and hold at every value of it. The fluctuation is the natural choice because it makes *large* mean what it means for the carrier itself: a change that moves the content by as much as the content moves of its own accord.

Method. The membrane is an Ornstein–Uhlenbeck process in units of its own stationary standard deviation, with the threshold at zero. After a spike

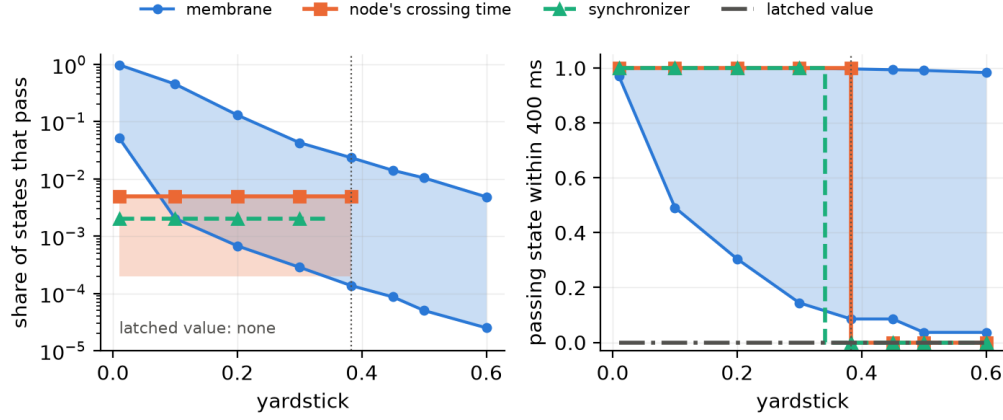


Figure 2: One protocol at every yardstick. Left: the share of occupied states at which every admissible change moves the next content by at least the yardstick. Right: the chance of entering such a state within 400 ms. The membrane is a band over every declared regime and, on the right, every noise share. A switching node's crossing time is a band over the declared activity factors; it passes exactly at the fluctuation (dotted) and at no higher yardstick. The synchronizer's share is the upper bound $T_W f_D$, at the yardsticks one bit can reach, up to $\Phi(1) - 1/2$. The value a clocked latch reads passes at no state and in no window.

it is refractory and then reset to its mean, and the mean is calibrated so that the stationary rate equals the declared rate. Probability is propagated on a voltage grid through the exact Ornstein–Uhlenbeck step, and a crossing between two sampled times is counted with the Brownian-bridge probability $\exp(-2xy/s^2)$ for sub-threshold samples x and y and step deviation s . Nothing is sampled. From each grid state, the law of the first spike time in the window, with its absence as one more outcome, is propagated in one pass, and the response to a change is the total-variation distance between two such laws. A state passes when every admissible change moves the law by at least the fluctuation, where a change is admissible if it is at least one noise amplitude of the carrier and does not carry the membrane across threshold. The fraction of occupied states that pass is the stationary mass of the passing

states. The probability of entering a passing state within a window is computed on the same chain with the passing states made absorbing. A carrier that supplies a share φ of the membrane’s noise variance has noise amplitude $\sqrt{\varphi}$ in the membrane’s units, and its changes are scaled accordingly. The bit is a carrier restored each step to a rail at a declared margin, read after one step of unit Gaussian noise, with occupancy given by that noise around the rail. The same computation is repeated at yardsticks from 0.01 to 0.6 in place of the fluctuation, at every regime and noise share. For correlated carriers, each membrane is the sum of a shared Ornstein–Uhlenbeck component of variance c and a private one of variance $1 - c$, both with the membrane’s time constant. Given the shared path the carriers are independent, so the probability that none of K passes within a window is the average of $(1 - h)^K$ over shared paths, where h is one carrier’s chance of passing given the path. For each of 256 shared paths from a fixed seed, the membrane’s probability is propagated through the exact step with the private noise, its centre moved by the shared path’s increment, after a burn-in of five time constants. Halving both the grid spacing and the time step changes the passing fraction by 10%, and the correlation of two carriers’ passing by at most 0.012.

Correlation lowers the per-window link. The chance that 100 carriers leave a window of 400 ms without a link is at most 0.0018 at $c = 0.25$, 0.017 at $c = 0.5$ and 0.12 at $c = 0.75$. These bounds, and the links of at least 0.88 and 0.98 in Section 3, lie two Monte Carlo standard errors on the conservative side of the estimate in every regime. The shorter window is more sensitive: within 40 ms the link falls to about 0.23 at $c = 0.75$, so P’s per-window condition is met at the time scale of a percept rather than of its fastest components.

The window is also swept on its own, from one time step to 1000 ms, for independent carriers. The chance that one carrier occupies a passing state at some step of a window is propagated once on the same chain, so every window comes from one pass, and the link of K carriers within it is $1 - (1 - h)^K$. The

link reaches one half in every regime and at every noise share from 3 ms with 1000 carriers and from 31 ms with 100, and since h grows with the window it stays above one half at every longer one. Halving the grid spacing and the time step moves both ends earlier, so these are the conservative values.

The switching chip is counted in closed form. On the rising half of a transition the node's voltage crosses its threshold at a slew $\dot{V}$, a change ΔV moves the crossing by $\Delta V/\dot{V}$ and the node's noise jitters it by $\sigma/\dot{V}$, so the response is $2\Phi(\Delta V/2\sigma) - 1$ whatever the slew. Every admissible change of at least one noise amplitude then reaches the fluctuation, exactly at one amplitude, so the node passes at the fluctuation and at no higher yardstick. A node that switches in a fraction α of the cycles of a clock of frequency f , with transitions of duration t , spends $\alpha ft/2$ of its time on the rising half of one, and a window of N cycles holds no transition with probability $(1 - \alpha)^N$. A synchronizer with aperture T_W , clock f_C and data changing at rate f_D samples inside its aperture at the rate $T_W f_C f_D$, so at most a share $T_W f_D$ of its samples do. Each sample is read as one bit, so it passes at no yardstick above $\Phi(1) - 1/2$, and at lower yardsticks that share bounds its passing fraction. A second stage allowed a time S to resolve fails on average once every $e^{S/\tau}/(T_W f_C f_D)$ seconds, with τ the stage's resolution time constant (Ginosar, 2011); the declared values are that tutorial's 28 nm example, $\tau = 10$ ps and $f_D = 100$ MHz, with S one cycle.

The one place a working chip latches a transition is a synchronizer between clock domains. With a 20 ps aperture and data changing at 100 MHz (Ginosar, 2011), its first stage samples inside the aperture at most 0.2% of the time, 8×10^5 times in a window of 400 ms, and which way it resolves moves with the timing of the transition; a second stage given one cycle to settle leaves it unresolved once in more than 10^{29} years. What that second stage takes is the bit the first resolves to. The noise in what the first stage samples includes the transition's own jitter, one amplitude of the change, and it is thermal, so Gaussian, and one bit under Gaussian noise does not reach

the fluctuation: at the centre of the aperture, where its two-sided response is largest, a change of one noise amplitude in the transition's timing moves the probability of either resolution by at most $\Phi(1) - 1/2 = 0.34$, and the stage's own noise only lowers that. The synchronizer therefore fails graded dependence at every sample, those inside its aperture included, whichever protocol carries the data across the boundary, and so does a random source that reads its bits from how a metastable stage resolves.

The ceiling depends on the noise law only through its mass near the centre. A bit read as 1 when a quantity plus noise exceeds a threshold responds to a change a by the noise law's mass on one of the two intervals of width a that meet at the threshold, so its two-sided response is the smaller of the two, at most half the mass of an interval of width $2a$. For a symmetric law with one peak the smaller mass is largest with the threshold at the state, where it is the mass between the centre and one amplitude. One bit therefore stays below the fluctuation at every threshold exactly when the law puts less than twice the fluctuation of its mass within one amplitude of its centre. Scaled to unit standard deviation, Gaussian, logistic and Laplace noise meet this, and a Student law with three degrees of freedom does not.

Several bits held near their thresholds together can read one graded quantity q , as the stages of a time-to-digital converter read one transition time. The word is a function of q plus the quantity's own noise and the stages' noise, and the stages' noise is independent of q . The response of the word to a change of q is therefore at most the response of q itself, $2\Phi(a/2\sigma) - 1$ under Gaussian noise of amplitude σ , by the data-processing inequality for total variation. With two thresholds half an amplitude on either side of the state and no stage noise, the word's three outcomes split the line where the two laws cross, and a change of one amplitude in either direction moves it by exactly the fluctuation. A word whose thresholds straddle the quantity within its noise thus passes exactly where the quantity passes.

G Digital hardware in detail

Clockless logic. Asynchronous circuits have no clock. In a quasi-delay-insensitive design, each stage signals completion to the next through a handshake, and correctness holds under every assignment of gate and wire delays apart from a few forks assumed isochronic (Martin & Nyström, 2006). Their transition times vary gradedly with every voltage, as a spike time does. But the design makes the computed values independent of those times: a value is accepted once its code is complete, whatever delays produced it. The timing of such a circuit is therefore a trace in the sense of Section 2. It moves with the physics, while the circuit’s own contents are read from completed codes, each held at a rail by restoring gates, so they meet the margin lemma as a clocked circuit’s contents do.

Leakage. Real digital hardware leaks: supply noise, crosstalk and temperature move the analog quantities behind every bit. The margin lemma says where such leakage can reach content. If a content read as a word of thresholded continuous quantities depends gradedly on any region in the exact sense, some bit of the next readout sits on its threshold. Correct sampled operation places those readouts inside margins. Gate restoration stabilizes the rails, and the network of gates supplies logical coordination. The constitutive graph tests whether an admissible physical change inside those margins reaches another content above the yardstick.

The margin also has a width. If every bit of the next state is at least δ from its threshold, each bit is read from an L_v -Lipschitz quantity and the microdynamics is L -Lipschitz, then a change to one region smaller than $\delta/(LL_v)$ changes no bit. Leakage below that scale reaches content only through the margin exception probability (Section 3). Above that scale the bound ceases to certify invariance; a sufficiently large disturbance can cause a logical error. Digital design keeps ordinary noise well below that scale: a textbook inverter’s static noise margin is 41% of its supply voltage, against an expected

supply variation of 10% (Rabaey et al., 2003), so the expected variation is 0.24 of the margin. The nondeterminism of floating-point reduction order is a choice among logical trajectories and lies outside both results: it changes which content is computed, not how content depends on graded state.

The margin’s width. A reliable rail and a near-threshold converter obey different response laws, even on one chip. Adding bits refines the number represented without narrowing any rail’s voltage margin. By the noisy margin theorem, reaching the fluctuation with an admissible change inside r/L requires a margin exception probability at least that large. Since that probability counts near approaches to thresholds, it is a different quantity from the processor’s logical error rate, and the criterion is applied to each physical readout over its actual input range. The examples compare rails 506 noise amplitudes from threshold with model membranes averaging 2.0 to 3.5 amplitudes below it.

A converter driven by an external graded signal is judged with that signal’s noise and its reader’s integration time, and a word of bits or repeated measurements may carry a graded response at that scale. For a hybrid system the question is which such carriers overlap, and whether their responses and restoration meet P in every content window.

H The bridge in detail

Instantaneous design estimates. For an additive Gaussian channel, at unit gain, an input shift of one source noise amplitude reaches the fluctuation when the effective channel noise is at most that amplitude. These quantities are evaluated at the chosen input and output coordinates, after filtering. A scalar quantizer suppresses a change at states sufficiently far from an edge. With the declared noise, being farther than the source’s noise plus 0.3 times the recording noise suffices for one admissible perturbation to miss the yard-

stick.

Under the information-matching approximation in the estimate, matching by adding noise to the analog channel would require noise of standard deviation 0.24 times the quantizer’s step. Keeping that channel sensitive then caps the step at 4.1 source noise amplitudes. Under uniform occupancy within a step, the resulting fraction of states certified to fail is at most 52%. Lowering bandwidth instead preserves this instantaneous noise budget while changing the time course. The reported step of 10 noise amplitudes, with the declared recording noise, gives a certified failing fraction of at least 74% under the same occupancy assumption.

A first-order filter of cutoff f_c has time constant $1/(2\pi f_c)$. The design floor of 1 Hz compares this constant with the restoration deadline in Appendix B. The preparation also measures total delay and the closed loop’s recovery of disagreement. The filter floor is a preliminary constraint; meeting the deadline is a property of the whole loop. Settings are fixed once the information controls are met and each arm’s temporal classification is established, and before any outcome is compared.

The complete reader and the content window. For a micro state x , let C_j^{next} be the next content actually taken from carrier j , including any integration of several samples by its reader. Let $\mathcal{A}_i(x)$ contain the admissible changes of source carrier i , between its noise and its threshold, with other coordinates fixed. Write

$$r_{ij}(x) = \inf_{s \in \mathcal{A}_i(x)} \text{TV}(\mathcal{L}(C_j^{\text{next}} \mid x_i \leftarrow s, x_{-i}), \mathcal{L}(C_j^{\text{next}} \mid x)).$$

The laws are interventional, using the same declared noise process in each case. An empty $\mathcal{A}_i(x)$ is excluded from certification. For a trajectory X and a content window T , define

$$G_{ab}(X, T) = \mathbf{1}\{\exists t \in T, i \in a, j \in b : r_{ij}(X_t) \geq \theta\}, \quad \theta = 2\Phi(1/2) - 1.$$

This expresses the carrier and window criterion used in Section 3. Both directions are evaluated, and the probability that both occur in the same window is estimated with their dependence retained. Restoring dynamics and its deadline are checked by a separate perturbation of overlap discrepancy.

The order of quantifiers matters. The instantaneous failing fraction measures time spent outside the passing set, whereas G_{ab} records whether the set is visited at all, and a brief approach to threshold is enough to set it to one. Several thresholds reading the same graded input can also preserve information that a single bit loses, as the example in Appendix F shows. A reader that integrates successive symbols is assessed on that joint sequence: repeated noisy observations can resolve an input change that a single observation misses. Filter memory, timing of code changes, recording noise correlations and downstream membrane integration belong in this next-readout law.

To exclude a route by the noisy margin theorem, apply it to the full map from the perturbed state to this next readout. Its Lipschitz constant and exception probability cover the whole map and all relevant samples, and an admissible change must remain inside its effective margin. A single-sample margin bound provides that certificate only when the next readout uses that sample alone. To establish a passing route, lower confidence bounds must cover every admissible change for a selected carrier pair; a finite grid needs justified monotonicity or regularity between its points. To establish failure, an upper bound below the yardstick for one admissible change suffices for each candidate pair, at every state the window visits. The observation window, intervention duration and tolerances are fixed before outcome comparisons. The preparation measures alternative anatomical routes as part of the same audit.

A specified dynamical alternative. Let z_t contain the recurrent observer’s estimates of the locations carried by the two regions. At the chosen

sampling interval its pilot-fitted update is

$$z_{t+1} = Az_t + Bu_{t-d} + K\ell_t + \xi_t,$$

where ℓ_t is local sensory evidence, u_t is the delivered bridge signal, d is its measured delay, and ξ_t has the calibrated residual law. The recorded input passes through the measured filters and, in the quantized arm, the quantizer before entering u_t . Parameters describing the cortical observer are shared across arms; the measured circuit transformation changes with bandwidth and resolution. The state includes filter and delay memory.

For the matching task let $m_t = w^\top z_t$ be the predicted difference of the two location estimates. If the decision is *same* when $|m_t + e_t| \leq r$, the model's choice probability is

$$p_{\text{same}}(t) = F_e(r - m_t) - F_e(-r - m_t),$$

with F_e the pilot-fitted continuous cumulative distribution of decision noise and a fixed criterion r . For unity reports the specified alternative is

$$\text{logit } p_U(t) = \gamma_0 + \gamma_D|m_t| + \gamma_V v_t + \gamma_q q_t,$$

where v_t is the observer's uncertainty about the location difference and q_t is the independently measured local visibility score. The coefficients are fitted on calibration blocks and fixed for the comparison. Both decision models use a common rule across arms, with no free coefficient for the analog or quantized label.

The factorial comparison fits the effects of bandwidth and quantization separately, while the resolution sweep at fixed bandwidth tests their interaction with noise and delay. The experiment records impulse responses, output spectra, stimulation gain and discrepancy recovery as well as transfer entropy and task information. Predictions concern the complete response and recov-

ery distributions on held-out blocks. This alternative can predict an analog advantage from smoothing or a quantized advantage from transmitted signal strength. P with R additionally rules out preserved unity reports in a condition certified to fail its coupling requirement. A condition that both accounts predict alike measures their common mechanism; a disagreement between their predictions defines the decisive comparison.

Whether the loops can be matched. The saved simulations assess information matching for transmission from region A to region B, at a 1 ms step and in units of A’s noise amplitude. The proposed preparation uses reciprocal loops and adds the temporal certification and observer comparison above. A task variable, a process of standard deviation 2 to 10 and time constant 20 or 100 ms, drives region A on top of A’s own noise. A is recorded with noise of its own amplitude and passed through the shared filter, then through the analog loop’s cutoff or the quantizer, and the shared output stage drives region B, a leaky integrator with noise of its own. Both quantities are estimated twice. The linear estimator takes transfer entropy from A’s recording to B’s by the linear-Gaussian formula, and task decodability as the variance of the task variable that a linear decoder of B’s recording explains on held-out trials. It misses what the quantizer’s nonlinearity passes on, so the nearest-neighbour estimator takes transfer entropy as a conditional mutual information (Kraskov et al., 2004; Frenzel & Pompe, 2007) and decodability as the variance a decoder would explain if the task information in B’s recording, estimated the same way, were Gaussian; on the analog loop the two estimators agree. For each step and estimator, two cutoffs are read off a grid from the floor above to 1 kHz: the one that matches the quantized loop’s transfer entropy and the one that matches its decodability. Both measures rise with the cutoff, so at the lower of the two the analog loop carries no more of either.

At the step of 10 noise amplitudes, in every regime whose task variable

spans at least half a step, the linear transfer entropy is matched between 17 and 95 Hz, well above the floor, and the matched analog loop explains 0.025 to 0.06 less of the task variable’s variance than the quantized loop. The same holds at every simulated step no larger than twice the task variable’s standard deviation. Lowering the bandwidth removes fast activity that carries transfer entropy and little of the task, while the quantizer removes small changes at every frequency, so at matched linear transfer entropy the quantized loop keeps more of the task. When the task variable spans only a fifth of a step, the matched analog loop can carry more task information, by up to 0.075, and the step has to be made finer before the loops are compared. The task is therefore chosen so that its variable spans several noise amplitudes of the recorded region.

By the nearest-neighbour estimator the quantized loop carries more transfer entropy than the analog loop does at any cutoff up to 1 kHz, because its jumps tell the driven region when the recorded one crossed a step, so at the declared step no cutoff matches it. At the steps where one does, the matched analog loop carries more task information in 4 of 7 cases, by up to 0.038: which side of a transfer-entropy match the task information falls on depends on the estimator. The lower match does not. The nearest-neighbour decodability is matched between 33 and 399 Hz, above the linear transfer-entropy match in every regime, so the lower match is the linear one, and a preparation sets the cutoff there. At that cutoff, by the nearest-neighbour estimator, the analog loop carries 10 to 55% less transfer entropy and explains 0.028 to 0.065 less of the task variable’s variance than the quantized loop, and it carries less of both at every simulated step no larger than twice the task variable’s standard deviation.

On the preparation, the decoder is fitted on other blocks of the same loop, the step and the cutoff are set together on pilot blocks, and the check is repeated on the blocks that are compared.

With blocks of 100 trials repeated on independent noise, the matched

transfer entropy and the gap vary from block to block, and their confidence intervals narrow with the square root of the trials. The behavioural comparison, accuracy 0.75 against 0.7 at a two-sided level of 0.05 and power 0.8, needs 1248 trials per loop, and the neural comparison, at a standardized difference of 0.2, needs 393. At the behavioural count, by either estimator, the model’s transfer entropies are estimated to within 4.4% and the difference in task variance to within 0.006, smaller than every margin above in these simulations. Those counts describe the declared pairwise effects. Pilot data determine the repeated-block variance, multiple-comparison plan and equivalence power for the factorial experiment and report readout. The limits of this transmission model are collected in Section 8.

The analog arm is kept analog end to end so that quantization is an explicit experimental factor. Each arm’s status under P is determined by the complete response protocol above.

Stimulation artefact. Event-triggered loops blank the recording around each pulse, and a continuous drive leaves no interval to blank. The loops therefore record and stimulate in separate frequency bands. Each loop reads multi-unit activity, above a few hundred hertz, takes its envelope, and drives the other region with a current that follows the envelope and so lies in the band below it. Rectifying and smoothing are analog operations, so the analog loop stays analog end to end. What leaks across the bands is removed by subtracting the loop’s own output, which the loop knows exactly, passed through the transfer from stimulator to recording contact measured on open-loop calibration blocks. Both loops end in the same output stage, which smooths the quantizer’s steps into the stimulation band, and share the stimulator, the bands and the subtraction, so both arms use the same artefact path. Their different waveforms can still produce different residuals. Open-loop blocks in which each stimulator follows activity recorded in another block estimate that residual separately in every bandwidth and quantization condition. Its

uncertainty is propagated to the response bounds and the outcome comparison.

I Timing the field’s cone

Brain tissue carries an electric field as well as synapses. At the frequencies of neural activity that field is quasi-static (Gratiy et al., 2017): it reaches every point of the tissue it spans with a propagation delay negligible on the neural time scales considered here. Synaptic influence between distant regions, by contrast, arrives only after axonal conduction and synaptic transmission, no sooner than 8 ms across the longest cortico-cortical fibres. If two regions re-agree after a local perturbation faster than the synaptic route allows, the coupling has a non-synaptic part, and the timing measures the reach of the kernel tail of Section 4 directly.

The field’s strength sets the distance at which the test of the field’s causal cone in Section 6 can decide anything. Endogenous fields in cortex peak near 2.4 mV/mm in vivo (Fröhlich & McCormick, 2010), networks of neurons detect fields from 0.14 mV/mm (Francis et al., 2003), and the potential of a local cortical source decays as $r^{-2.1}$ across millimetres (Rebollo et al., 2021). With the peak placed at the nearest distance of that measurement, the field of a local source falls to the detection threshold within 3.8 mm, and misses it by 5 orders of magnitude at 15 cm and by 3.5 at 5 cm. The field’s cone is therefore tested at millimetre distances, where activity has been observed to synchronize across a cut through cortex by the field alone (Rebollo et al., 2021). At centimetre distances P predicts that re-agreement arrives through synaptic coupling, which is graded as well.

The timing can be measured with intracortical recording, with care. The stimulation artefact saturates clinical amplifiers during the first milliseconds, masking the earliest monosynaptic responses (Keller et al., 2014), and early responses at distant contacts carry a volume-conducted component that

scales with distance as the artefact does (Prime et al., 2020). Three measures separate active re-agreement from the passive field that carries the stimulus: template subtraction of the artefact (Trebaul et al., 2016); multi-unit activity or current-source density at the distant site, which the passive spread of the local field potential does not reach (Kajikawa & Schroeder, 2011); and a deadline set for each subject from the tractography length of the connecting fibres (Caminiti et al., 2013).

J Further related work

Embodied architectures. Starzyk & Galus (2026) propose that candidate conditions for consciousness involve sensory representations integrated with regulation, affect and recurrent feedback in the Motivated Emotional Mind (MEM) architecture, and leave realization in artificial substrates as a hypothesis to be tested through causal organization and temporal dynamics. P gives one part of that question, the coupling of contents, a mechanistic criterion: whether their carriers respond gradedly above noise and restore agreement within a content window, measured by the same intervention on any substrate. MEM’s call for comparisons with competing architectures and hybrids is the kind of explicit model comparison the bridge builds in.

Biology and the medium. Seth (2026) argues that consciousness may depend on biological properties that computation does not capture. P makes one such property explicit, and measurable: sensitivity above noise and restoration within a window, assessed by one protocol on membranes and logic nodes. It locates that property in how the substrate is organised, and admits non-biological hardware that has it (Section 5). Mortal computation, whose parameters are useful only on the hardware instance that learned them (Hinton, 2022; Ororbia & Friston, 2023), shares P’s refusal to separate the computation from the medium.

The distinction also concerns where a learning signal is available. Under the noisy-margin hypotheses a content read from bits has a response no larger than its margin exception probability (Section 3; Appendix K); a graded substrate can carry a local response to a weak nudge as it relaxes. Weak nudges followed by relaxation are also used in equilibrium propagation (Scellier & Bengio, 2017), while physical networks can be trained with a hybrid physical and digital backpropagation procedure (Wright et al., 2022). P supplies neither a learning rule nor an advantage in learning or representation: digital hardware can retain smaller changes by using more bits, and the reach bound in Section 4 expresses consistency across a distributed content.

Analog and digital. On the mechanistic account of Piccinini (2008), a digital computer manipulates strings of digits, states its components distinguish reliably, and which class a system belongs to is fixed by its mechanism. P’s symbol, a content read through a noise margin, is that reliability stated as a property of the readout, with the margin measured in the parts’ own noise; P then asks what the reliability costs, namely graded dependence below the margin. Maley (2011) separates analog from continuous and digital from discrete representation, and P’s line follows the second pair, not the first: a spike time is a physical content whether or not it represents by analogy, and a continuous quantity latched through a margin is a symbol. Maudlin (1989) argues that a computation’s counterfactual structure can be carried by machinery that is idle on the actual run, which sits badly with consciousness supervening on present physical activity. P states its counterfactual condition on present activity: graded dependence asks how the actual carriers respond to changes of the actual state, above their noise, and idle machinery responds to none.

Field theories. The electromagnetic field theory of McFadden (2020) identifies consciousness with the brain’s field, supported by evidence that endogenous fields entrain cortical activity (Fröhlich & McCormick, 2010). Miller et

al. (2026) place cognition and consciousness in travelling waves that synaptic and ephaptic coupling sustain together, and take consciousness to arise when those waves bring cortex into a globally integrated state. Their analog computation is set apart from digital computation by running in parallel. P treats the field, and the waves it carries, as ways among several to realize graded coupling, and draws the line between analog and digital at the noise margin, which a parallel digital machine keeps.

K Formal results

Table 1 lists the machine-checked results used here, from the Lean 4 development in the project repository, in its four **Phase10** modules. Table 1 maps each result to its declaration and to the section that argues from it. Every theorem rests only on Lean’s foundational axioms **propext**, **Classical.choice** and **Quot.sound**, which the build verifies for every declaration of the development. Small concrete models in **Examples/** meet the hypotheses of the bounds and show where they are attained. In this table, a system means a component of the stipulated coupling graph. The digital exclusion rows use the full noisy-margin hypotheses, including an admissible change above noise within the effective margin for every source. Connectivity and the probabilistic reach bounds concern sensitivity; contraction and the exponential interpretation are distinct requirements. The observer equations in Appendix H specify an experimental model.

Table 1: The paper’s formal results and the Lean declarations that carry them.

Result	Section	Declaration
A readout with a margin, under continuous micro dynamics, has no graded dependence on any region	3	not_physicallyEnforced_of_margin

Result	Section	Declaration
The same for any content read from a finite word of thresholded continuous quantities	3	<code>not_physicallyEnforced_of_bits</code>
A thresholded continuous quantity has a margin off its threshold	3	<code>hasMargin_threshold</code>
A category read from a carrier depends gradedly on a region only where the carrier does	3	<code>GradedDependence_of_comp</code>
Every function of a bit word, after steps that keep the bits off their thresholds, depends gradedly on no region	3	<code>not_gradedDependence_iterate_of_bits</code>
Shells in series bound the effective conductance	A	<code>conductance_le_shell</code>
Crossing coupling at most $c(k+1)$ gives resistance at least $\log(n+1)/c$	A	<code>conductance_mul_log_le</code>
An averaging chain from a to b bounds the resistance by $2(\sum_k \rho_k)^2$	A	<code>one_le_conductance_of_chain</code>
A tail $K(r) \gtrsim r^{-(2+\sigma)}$ with $\sigma < 2$ bounds the conductance below at every distance	A	<code>le_conductance_of_powerLaw</code>
Adding coupling never raises the effective resistance	A	<code>conductance_mono</code>
A direct coupling caps the resistance at its reciprocal	A	<code>le_conductance_of_edge</code>
A Lipschitz encoder carries mean-square phase agreement to content agreement	4	<code>integral_dist_sq_le</code>
A relative encoder reports one content at every site of a wave	A	<code>relContent_eq_of_wave</code>

Result	Section	Declaration
A relative encoder's discrepancy is bounded by the departure from a wave	A	<code>dist_relContent_le</code>
Tracking a remote variable requires it to lie in the causal past	4	<code>mem_ball_of_reconstructs</code>
Contraction by κ reaches ε by step $\log(d_0/\varepsilon)/\log(1/\kappa)$	4	<code>disc_iterate_le_of_log_le</code>
A discrepancy shrinking by at most ρ per step is above ε before $\log(d_0/\varepsilon)/\log(1/\rho)$	4	<code>lt_disc_iterate_of_lt_log</code>
Graded dependence of a bit word requires some bit exactly on its threshold	5	<code>exists_threshold_of_gradedDependence</code>
A one-region change below $\delta/(LL_v)$ changes no bit	5	<code>bits_eq_of_lipschitz</code>
Under noise, a one-region change below r/L moves the probability of every event of the next content by at most the margin exception probability	3	<code>law_le_of_marginRate</code>
The same for anything computed from noisy bits, with margin exception probability the probability of landing within δ of a threshold	3	<code>bits_law_le_of_lipschitz</code>
The response of a content to one region grows with the scale	3	<code>response_mono</code>
Below r/L the response is at most the margin exception probability	3	<code>response_le_of_marginRate</code>
Below r/L the finite-difference nudge signal is at most the margin exception probability divided by the nudge size	J	<code>nudgeSignal_le_of_marginRate</code>

Result	Section	Declaration
For symmetric overlaps, graded dependence at every directed overlap holds exactly when every overlapping pair is directly coupled	3	<code>enforcedAtNoiseFloor_iff</code>
A region whose content has a margin with margin exception probability below the fluctuation is coupled to no region, and is one system only with itself	3	<code>not_coupledAtNoiseFloor_of_record, sameSystem_iff_eq_of_record</code>
For any set of such records, two regions are one system exactly when a chain of couplings avoiding every record joins them	3	<code>sameSystem_iff_of_records</code>
A content that depends on a region only through a record's symbol responds to it by at most the record's margin exception probability, so write-back through a record couples nothing	3	<code>shiftResponse_le_of_factors, not_coupledAtNoiseFloor_of_factors</code>
A record that reads two graded regions joins neither, a write-back through it links neither, and a graded reader under linear dynamics with uniform noise joins two uncoupled regions	3	<code>tally_sameSystem_iff, writeBack_not_coupled, reader_sameSystem</code>
Asking graded dependence above the noise floor of every carrier implies asking it of some carriers of each region	3	<code>EnforcedAtNoiseFloor.enforcedByCarriers</code>

Result	Section	Declaration
A state at which some carriers of each region pass and not every carrier does	3	<code>spare_</code> <code>enforcedByCarriers,</code> <code>spare_not_</code> <code>enforcedAtNoiseFloor</code>
Regions joined by carriers at every overlap are one system	3	<code>EnforcedByCarriers.</code> <code>regionSameSystem</code>
Bits with margin exception probability below the fluctuation: no carrier of any region reaches any other, and each region is a system of its own	3	<code>not_</code> <code>reachesAtNoiseFloor_of_</code> <code>bits, regionSameSystem_</code> <code>iff_eq_of_bits</code>
With K independent candidate carriers each giving directed reach within a window with probability at least p , that direction fails with probability at most $(1 - p)^K$	3	<code>measure_not_</code> <code>reachesWithin_le</code>
Without independence, directed reach fails with probability at most the variance of the number of passing carriers over the square of its mean	3	<code>measure_not_</code> <code>reachesWithin_le_</code> <code>variance</code>
Bits with margin exception probability below the fluctuation couple no pair of regions, so each region is a system of its own	3	<code>sameSystem_iff_eq_of_</code> <code>bits</code>
The same bits fail graded dependence above the noise floor at any overlap of two regions	3	<code>not_</code> <code>enforcedAtNoiseFloor_</code> <code>of_bits</code>
A smaller shift of a Gaussian moves no event more than a larger one	3	<code>gaussian_shift_le</code>

Result	Section	Declaration
Where the response grows with the change, graded dependence above the noise floor is decided by a change of one noise amplitude	3	<code>gradedAboveNoise_iff_of_mono</code>
Under linear dynamics with Gaussian noise, the response grows with the change, and graded dependence and coupling are decided at one noise amplitude	3	<code>gaussLin_coupledAtNoiseFloor_iff</code>
A shift of a density with a single peak moves no event more than a larger shift, in either direction	3	<code>lawRise_map_add_mono_of_unimodal</code>
Under linear dynamics with additive noise whose density has a single peak, graded dependence and coupling are decided at one noise amplitude	3	<code>noisyLin_coupledAtNoiseFloor_iff</code>
Laws with a monotone likelihood ratio, exponential families among them, cross once	C	<code>singleCrossing_of_mlr,</code> <code>singleCrossing_expFamily</code>
Where the next content's laws cross once along a parameter the region moves monotonically, graded dependence above the noise floor is decided by a change of one noise amplitude up and one down	C	<code>gradedAboveNoise_iff_of_crossing</code>